

# Vulnerability Assessment Report

30<sup>th</sup> July 2026

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## System Description

The server hardware consists of a powerful CPU processor and 128GB of memory. It runs on the latest version of Linux operating system and hosts a MySQL database management system. It is configured with a stable network connection using IPv4 addresses and interacts with other servers on the network. Security measures include SSL/TLS encrypted connections.

## Scope

The scope of this vulnerability assessment relates to the current access controls of the system. The assessment will cover a period of three months, from May 2026 to July 2026. [NIST SP 800-30 Rev. 1](#) is used to guide the risk analysis of the information system.

## Purpose

The purpose of this vulnerability assessment is due to the risk associated with the current access controls, the database has been public for over 3 years. The database server is a key functional asset to the business where customer and business data are stored. It is critical to secure the system because of its regular use for marketing operations and to avoid sensitive data such as PII and SPII from leaking or exploited by a public threat actor.

## Risk Assessment

| Threat source | Threat event                                  | Likelihood | Severity | Risk |
|---------------|-----------------------------------------------|------------|----------|------|
| Competitor    | Obtain sensitive information via exfiltration | 2          | 3        | 6    |
| Employee      | Alter/Delete critical information             | 1          | 3        | 3    |
| Hacker        | Conduct Denial of Service (DoS) attacks.      | 3          | 3        | 9    |

## **Approach**

Risks considered the data storage and management methods of the business. The likelihood of a threat occurrence and the impact of these potential events were weighed against the risks to day-to-day operational needs.

## **Remediation Strategy**

Implementation of authentication, authorization, and auditing mechanisms to ensure that only authorized users access the database server. This includes using strong passwords, role-based access controls, and multi-factor authentication to limit user privileges. Encryption of data in motion using TLS instead of SSL. IP allow-listing to corporate offices to prevent random users from the internet from connecting to the database.